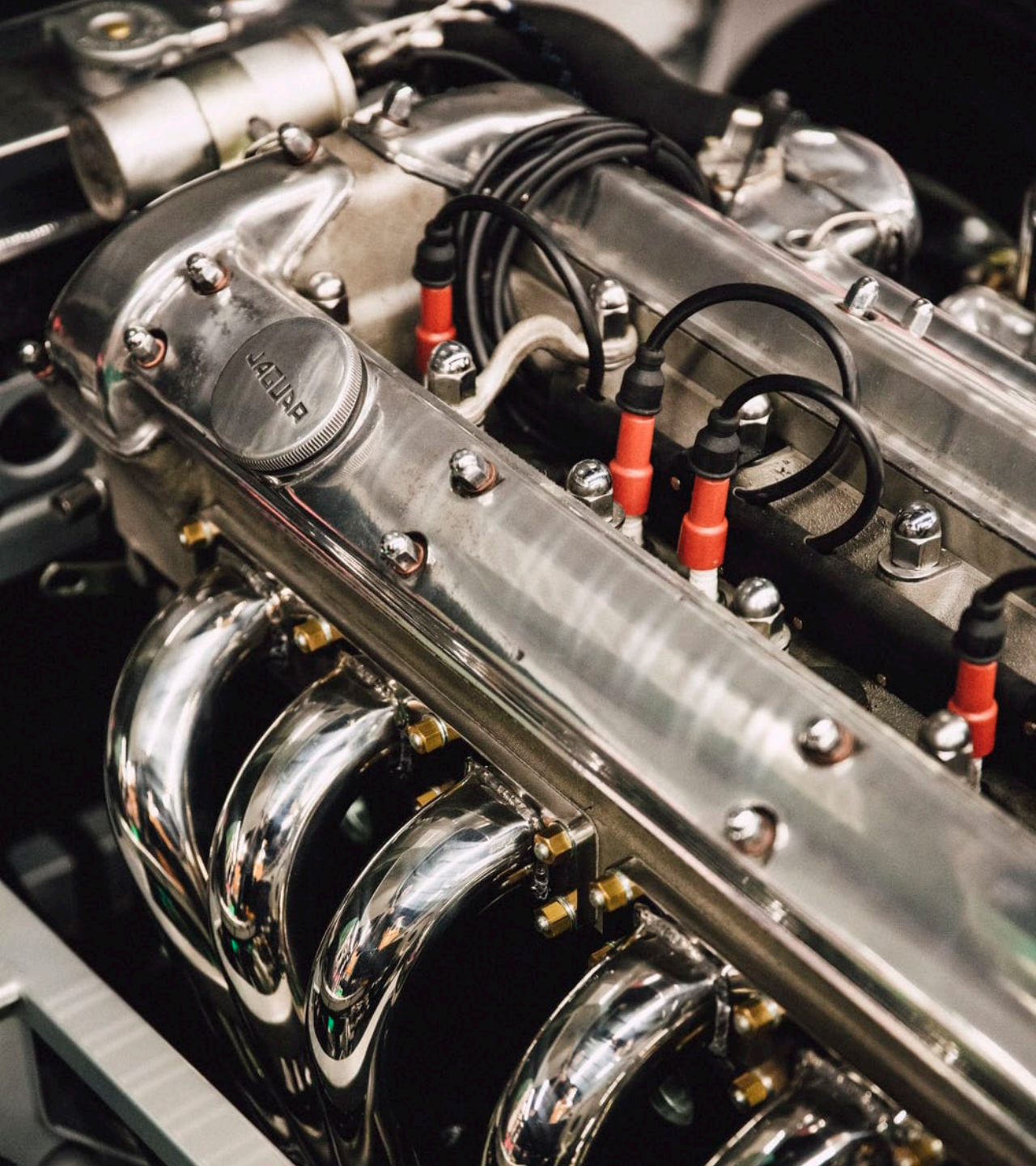
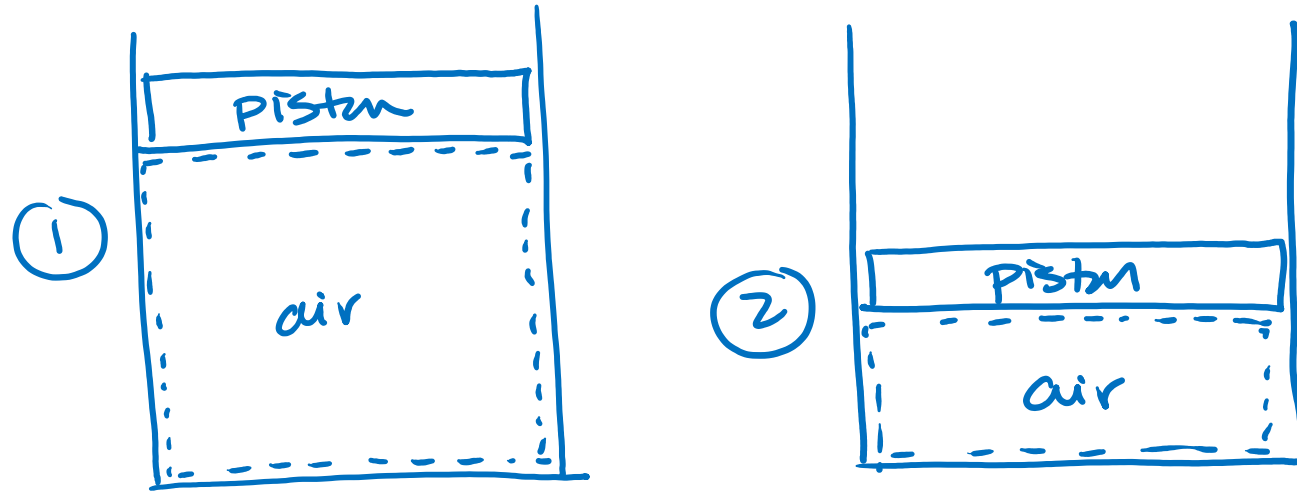




ME 300: Engineering Thermodynamics

Example: Processes and Cycles

Piston-cylinder process #1



$$\begin{aligned} P_1 &= 0.1 \text{ MPa} & P_2 &= 0.1 \text{ MPa} \\ V_1 &= 1 \text{ m}^3 & V_2 &= 0.5 \text{ m}^3 \\ M &= 1 \text{ kg} & M &= 1 \text{ kg} \end{aligned}$$

Find: T_1 T_2 ?

Assume

ideal gas

$$\rightarrow R = 287 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$



What kind of process is this?

- a) Isothermal
- b) Isobaric
- c) Isochoric

Piston-cylinder process #1

State 1

$$P_1 V_1 = M R T_1$$

$$T_1 = \frac{P_1 V_1}{M R}$$

$$= \frac{(0.1 \times 10^6)(1)}{(1)(287)}$$

$$T_1 = 348 \text{ K}$$

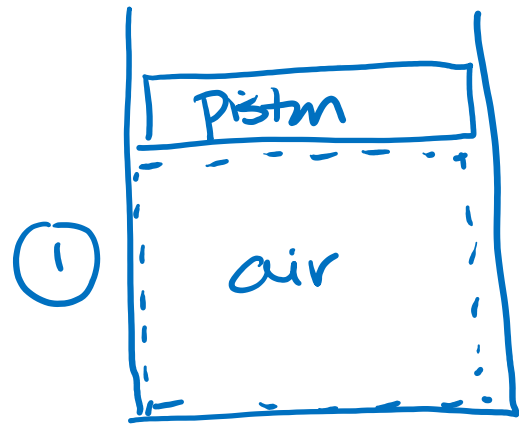
State 2

$$T_2 = \frac{P_2 V_2}{M R}$$

$$= \frac{(0.1 \times 10^6)(0.5)}{(1)(287)}$$

$$T_2 = 174 \text{ K}$$

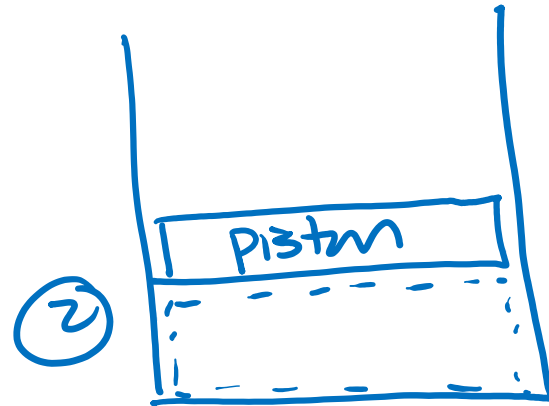
Piston-cylinder process #2



$$P_1 = 0.1 \text{ MPa}$$

$$V_1 = 1 \text{ m}^3$$

$$M = 1 \text{ kg}$$



$$P_2 = 0.2 \text{ MPa}$$

$$V_2 = 0.5 \text{ m}^3$$

$$M = 1 \text{ kg}$$

Find: T_1, T_2

Assume: ideal gas

Piston-cylinder process #2

State 1

$$T_1 = \frac{P_1 V_1}{MR}$$

$$= \frac{(0.1 \times 10^6)(1)}{(1)(287)}$$

$T_1 = 348 \text{ K}$

State 2

$$T_2 = \frac{P_2 V_2}{MR}$$

$$= \frac{(0.2 \times 10^6)(0.5)}{(1)(287)}$$

$T_2 = 348 \text{ K}$



What kind of process is this?

- a) Isothermal
- b) Isobaric
- c) Isochoric